

Yiğit Sarp Avcı

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Education

Boğaziçi University - B.Sc. in Computer Engineering

09/2023 – Expected 06/2028

- Extracurriculars: Executive Board Member & DJ at Radyo Boğaziçi, Regional Organizational Team Member at ESTIEM; Student-Athlete (Sailing, Skiing & Windsurfing), gave educational mentorship for high school students, participated in Work and Travel USA; Volunteering Division Member at Industrial Engineering Club (BÜYAK).
- Tools, AI & Ecosystem: Linux/Unix, Git/GitHub, AI-Assisted Development, ccache, Makefile, LaTeX, MS Office

Kabataş Boys' High School

09/2018 – 06/2023

- Graduated with a GPA of 3.78/4.00; Extracurriculars: Debate, Model United Nations, Entrepreneurship, and Coding Club; Volunteer: Community Involvement Projects and TOÇEV (Tüvana Okuma İstekli Çocuk Eğitim Vakfı).

Experience

Backend Developer Intern at DefineX

09/2026 – Present

- Joining the backend development team to contribute to the design and implementation of scalable software solutions. Excited to take on responsibilities involving robust backend architecture, API development, and system optimizations in a dynamic enterprise environment.

Site Reliability Engineering Summer Intern at Akbank

07/2026 – 09/2026

- Completed a 2-month summer internship with the SRE team, contributing to the reliability, scalability, and performance of mission-critical banking systems. Took on responsibilities involving infrastructure operations and systems engineering in a large-scale enterprise environment.

Software Engineering Summer Intern at Smash Mate

06/2026 – 07/2026

- Completed an intensive 1-month summer internship during the critical pre-launch phase, contributing to a complex microservices-oriented backend (Java, Quarkus) and a mobile frontend application (React Native). Developed core backend features, resolved high-priority issues, and authored comprehensive unit/integration tests to ensure zero-defect deployments. Played an active role in agile workflows, code reviews, and system optimizations, leading directly to the product's successful official launch.

Technical Projects (@GitHub)

FinTech & Trading Systems — Python, Next.js, FastAPI, PostgreSQL, Alembic, Pytest, WebSockets, Docker

- Developed **AlphaTrigger** (a real-time trading simulator featuring a Z-Score mean reversion strategy and volatility-adjusted position sizing) and the **FinTech Intelligence Platform** (featuring a live limit-order matching engine, portfolio risk analyzer, and AI-powered insights).

Enterprise Dashboards & Fullstack — Next.js 15+, TypeScript, Express, Prisma, TypeORM, Tailwind CSS, SWR

- Developed **Performo**, a real-time administrative dashboard with role-based access control, task management, and optimized database queries, alongside an **Enterprise FAQ System** featuring SWR-powered optimistic UI updates, Express backend, and robust API error boundaries.

Systems & CPU Scheduling — x86-64 Assembly, GNU Toolchain, Linux Syscalls, Process Scheduling, Multithreading (Pthreads), Memory Management

- Implemented **SchedSim**, a CPU scheduling simulator written in x86-64 GNU Assembly using direct Linux syscalls to execute FCFS, SJF, SRTF, Priority, and Round Robin scheduling algorithms.

Interpreters & Language Design — C, Python, Bash, Lexer Design, Functional Paradigms

- Built **Spirelayer** and **Aura**, including a C-based command interpreter with custom backtracking lexer/parser and a Python functional-language interpreter with closures, higher-order functions, and lexical/dynamic scoping.

Algorithms & Matching Engines — Java, Graph Theory, Heaps, Complexity Analysis

- Developed **MatrixNet** (a network graph analysis tool implementing Tarjan's SCC algorithm and articulation point detection) and **GigMatch-Pro** (a candidate matching engine using max-heap data structures for priority-based job allocation).

Mobile Development, Game & WebAssembly Engineering — React Native, Expo, C++, Qt6, WebAssembly, Java, TypeScript, Reanimated, Audio API, CI/CD

- Built **Smash Minesweeper** (a polished mobile app featuring a reducer-driven state engine, BFS flood-fill, and custom audio effects), **Qt 2048** (a C++/Qt6 clone cross-compiled to WebAssembly with undo history), and **Nightpass** (a terminal-based survival card game with a state-machine rule engine).